

Ian Schillebeeckx

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EDUCATION

Washington University in St. Louis

PhD, Computer Science (Machine Learning) – Fastest graduation in department history

St. Louis, MO

2012 – 2016

Saint Louis University

Honors BS, Computer Science

Honors BS, Applied Mathematics

St. Louis, MO

2007 – 2011

EXPERIENCE

Sr. Director of Enterprise Data Science and AI

CareDx, Inc. (Transplant Diagnostics)

January 2025 – Present

San Francisco, CA

- Established company's first data science function from zero, defining the rigorous analytical frameworks that measured clinical and commercial impact; served as primary quantitative consultant to CEO, CFO, and COO, translating complex statistical findings into compelling strategic narratives that influenced enterprise-level investment decisions
- Defined and directed causal inference research methodology: built causal models (DAGs, causal random forests) quantifying the marginal impact of operational interventions on clinical and revenue outcomes; established technical "source of truth" through controlled pre-post validation with measurable business impact (10% revenue increase)
- Architected multi-year research roadmap: built forecasting capability from scratch (~75% of revenue) with custom time series optimization reducing error from 6% to 0.9%; scaled from manual analysis to automated production system with self-service interface, monitoring, and 90-day revalidation
- Built predictive clinical risk identification system covering \$270M revenue: novel piecewise linear + dynamic programming approach for forward-looking prediction; deployed automated anomaly detection enabling proactive intervention on at-risk accounts across all US transplant clinics
- Drove AI transformation: deployed GenAI RAG application company-wide (\$1M investment); built agentic paper generation application producing 12 research manuscripts (2 submitted to journals); pioneered agentic workflows improving team productivity 50%; authored enterprise Responsible AI policy
- Led company-wide AI adoption to 92% (~750 employees), saving 8% of total hours annually; coached ~20 data science practitioners through regular training sessions on rigorous methodology

Sr. Director of Data Science

CareDx, Inc.

January 2024 – January 2025

San Francisco, CA

- Led Data Science organization (14 people, \$6M budget) spanning Bioinformatics, Statistics, and Data Science; hired, mentored, and developed senior individual contributors into technical leaders; fostered culture of scientific rigor and continuous learning
- Directed development of LSTM deep learning model for sequential clinical prediction from longitudinal patient data (~1200 patients, ~5000 time points), outperforming Kalman Filters, HMMs, and logistic regression; presented as oral abstract at international transplant conference (top 10% of submissions)
- Directed two production ML models (SVM, LASSO) deployed in regulated clinical environment with automated regression testing, model monitoring, and CI/CD; wrote regulatory filings (CLIA, NY-state)
- Pioneered first evidence demonstrating test impact on therapeutic decision-making through interventional study design; expanded DS influence into cross-functional operations through causal observational studies

Head of Product and Data Science

Cofactor Genomics (Oncology Diagnostics)

December 2017 – January 2024

San Francisco, CA

- Owned end-to-end delivery of 2 production ML models in CAP/CLIA regulated environment: defined research frameworks, led biomarker discovery across multiple cancer types, designed clinical and analytical performance metrics, deployed scalable cloud architecture with automated model monitoring
- Designed and executed 3 pre-specified clinical validation studies across 12+ sites (~500 subjects): statistical power analysis, prespecified hypotheses, rigorous analysis achieving Medicare reimbursement approval
- Led multidisciplinary team of 10; defined multi-year research and product roadmaps; translated complex clinical findings into strategic narratives for investors, partners, and key opinion leaders through 30+ physician interviews
- First author on Nature ML paper (2nd most downloaded in 2022); authored 3 AI patents (2 granted); led multi-center publication (PREDAPT study, PMC 2024)

CEO & Founder

Where4Care (Healthcare Technology Startup)

May 2017 – August 2020

St. Louis, MO

- Built healthcare product from zero in startup environment: designed patient matching algorithm, HIPAA-compliant tech stack (Python/Django); validated through 200-patient clinical study; scaled from manual research to automated product

CEO / Entrepreneur in Residence

June 2014 – May 2017

Sling Health (Healthcare Startup Incubator)

St. Louis, MO

- Co-founded innovation accelerator; mentored 50+ healthcare technology projects from problem definition through launch; scaled to 130+ participants; drove 200% revenue growth over 2 years

SELECTED PUBLICATIONS & PATENTS

3 Artificial Intelligence Patents (2 granted) – Methods for prediction, classification, and data processing using ML/AI
ML-based Biomarker – *Nature Scientific Reports, 2022* (2nd most downloaded paper that year)

12 Peer-Reviewed Publications – Clinical analytics, applied ML, causal methods (*CVPR, ECCV, Nature, J. Mol. Dx, PMC*)

TECHNICAL SKILLS

Clinical & Causal Methods: Causal Inference (DAGs, Causal Forests), Clinical Risk Identification, Interventional & Observational Study Design, Statistical Power Analysis, A/B Testing, Hypothesis Testing, Regulatory Filings (CLIA, CAP)

ML & AI: Supervised/Unsupervised Learning, Deep Learning (LSTM), Predictive Modeling (Forecasting, Risk Prediction, Classification, Regression), GenAI (LLM/RAG), Agentic AI, Production ML (SageMaker, CI/CD, Monitoring)

Languages & Platform: Python (expert), SQL (expert), R, AWS (SageMaker, Batch), Databricks, Tableau, Streamlit, Docker, Git

Leadership: 0-to-1 Data Science Builder (up to 14 reports, \$6M budget), Executive Influence & Narrative Building, Clinical Research Leadership, Peer-Reviewed Publishing, Mentoring Senior ICs into Leaders